

PROFESSIONAL EXPERIENCE

Goldman Sachs

August 2025 - Present

Associate Software Engineer (SDE2) - Liquidity Risk Applications

- Architected mission-critical regulatory compliance platforms processing vast financial datasets to compute FR2052a and other Federal Reserve Board mandated liquidity risk metrics with stringent accuracy standards
- Led cloud migration initiative, designing scalable AWS infrastructure utilizing ECS, Fargate for serverless container orchestration, ECR for container management, S3 for secure storage, and MWAA for orchestrating complex data pipelines
- Engineered sophisticated multi-stage Apache Airflow DAGs in Python to orchestrate end-to-end risk calculation workflows including data extraction, transformation, validation, and reporting with comprehensive error handling
- Spearheaded an advanced Generative AI analytics platform utilizing multi-agent LLM systems to automate dynamic regulatory report generation, leveraging AWS Bedrock, SageMaker, AgentCore, PyTorch, LangChain, and Google's Agent Development Kit (ADK)
- Designed custom LLM architectures fine-tuned on financial risk domain datasets, implementing agentic orchestration with AWS AgentCore and LangChain, coordinating specialized AI agents for data extraction, quantitative analysis, narrative generation, and compliance checking
- Ensured platform reliability and regulatory compliance through high-availability architectures, disaster recovery planning, and security controls meeting federal banking requirements

WissenIT

February 2025 - August 2025

Software Engineer Intern

- Built and maintained Spring Boot microservices to store and deliver patient medical records for internal hospital use
- Developed RESTful APIs to securely transmit EMR data to doctors and front-desk staff, ensuring HIPAA compliance
- Designed React-based frontend components for accessing and recording patient data
- Deployed services using Docker and Kubernetes to ensure high availability and scalability

NCR Voyix

April 2023 - August 2023

Software Engineer II

- Architected and deployed 25+ microservices for an internet banking platform serving 400+ credit unions, scaling to 5M+ transactions daily
- Engineered a scalable LLM-powered Virtual Banking Assistant, fine-tuned on customer financial data to analyze spending behavior, assist with transaction searches, and provide personalized financial insights.
- Orchestrated the end-to-end design of microservices architecture, integrating with financial institutions like Fiserv, and Zelle for seamless banking operations
- Tuned high-performance Java Spring WebFlux microservices, leveraging Netty reactive servers, achieving 99.99% uptime and 40% reduced latency
- Optimized distributed caching strategies with Redis and PostgreSQL, reducing response time by 35% for high-traffic banking requests
- Streamlined CI/CD pipelines using GitHub Actions, ArgoCD, and Helm, ensuring smooth version-controlled Kubernetes deployments on Google Kubernetes Engine (GKE)
- Hardened platform security & scalability by integrating SonarQube, AquaSec, and Mend, proactively eliminating 90%+ vulnerabilities, while load testing with JMeter to optimize performance, scaling deployments to support 2M+ users per bank across 400+ banks

NCR Voyix

August 2021 - April 2023

Software Engineer I

- Developed a Central Authentication & Configuration Service for secure multi-bank authentication using scalable JWT sessions
- Programmed adapter microservices for secure transaction routing & banking data exchange with third-party vendors
- Containerized & deployed microservices with Docker & Kubernetes, integrating Helm charts for efficient release management
- Built event-driven architectures using Apache Kafka, processing real-time banking transactions with low latency
- Developed RESTful APIs with Spring WebFlux & PostgreSQL, enhancing data access speeds by 50%

NCR Voyix

February 2021 - August 2021

Software Engineer Intern

- Automated batch job execution for credit card reward processing using Google Cloud Scheduler, Apache Beam, and DataFlow
- Prototyped a scalable rewards extraction engine using Google Dataproc, enhancing data batch processing reliability

Virtusa

June 2020 - November 2020

Software Engineer Intern

- Delivered a machine learning-powered IoT automation system managing 1,000+ interactions daily, cutting manual intervention by 40%

Technical Skills

Languages: Python, Java, Kotlin, JavaScript, Golang, C, C++, Bash

Frameworks and libraries: Spring Boot, Spring WebFlux, Spring Batch, React, Redux, Node.js, Angular

Databases: Snowflake, PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Cassandra, Bigtable

Cloud & DevOps: Docker, Kubernetes, Helm, ArgoCD, Terraform, CI/CD (GitHub Actions, Jenkins), Git, AWS (ECS, Fargate, ECR, S3, SageMaker, Bedrock, AgentCore, MWAA), GCP, Apache Kafka, Apache Airflow, Google PubSub

Machine Learning & NLP: PyTorch, TensorFlow, Transformers, LangChain, AWS SageMaker, AWS Bedrock, AWS AgentCore, Google Agent Development Kit (ADK), LLM Fine-tuning, Multi-Agent Systems

Monitoring & Security Tools: AppDynamics, Splunk, Google Cloud Monitoring, AWS CloudWatch, SonarQube, AquaSec, Mend

Education

University of North Carolina at Charlotte <i>Master of Science in Computer Science (GPA: 3.90 / 4.00)</i>	August 2023 - December 2024
Jawaharlal Nehru Technological University <i>Bachelor of Technology in Computer Science (GPA: 8.2 / 10.00)</i>	June 2017 - July 2021

Projects

- Real-Time Transaction Analytics Platform**
 - Designed a platform to analyze high-velocity transaction data using Kafka and Python, enhancing fraud detection capabilities by 40%
 - Built interactive dashboards with React, enabling financial institutions to monitor real-time transaction trends.
- Intelligent Credit Risk Assessment System**
 - Implemented a machine learning model for dynamic credit risk evaluation, leveraging TensorFlow and scikit-learn to improve accuracy by 30%.
 - Integrated APIs to securely fetch and process financial data, ensuring compliance with industry standards.
- Predictive Account Monitoring Tool**
 - Developed a tool to forecast unusual account activity, using anomaly detection algorithms in Python.
 - Automated reporting pipelines with Google Cloud DataFlow, ensuring real-time alerts for flagged transactions.

Coursework

Software Engineering, Cloud Computing, Intelligent Systems, Advanced Database Management Systems, Machine Learning, Networking, Data Structures, Algorithms, Full-Stack Application Development, Object Oriented Design with Java, Software System Design Patterns